Overview: Universal Constants and Curvature as Projection Phenomena

Document Register: Proof of Concept (PoC) Foundation
Classification: Technomouse and the genie of the bottle (Register 2)

This document establishes the macro-theoretical framework for redefining the fundamental constants of the universe and the nature of spacetime curvature. Rather than accepting these values as arbitrary "givens," this model postulates they are direct consequences of projecting a 7-dimensional manifold (T^7) into 4-dimensional Minkowski spacetime (M^4).

1. Constants as Geometric Scaling Ratios

In standard physics, values such as the speed of light (*c*), Planck's constant (*h*), and the fine-structure constant (*α*) are treated as fundamental but arbitrary fine-tuned numbers. Under the $T^7 \rightarrow M^4$ projection model, these constants are redefined as the **structural tolerances of the projection**.

They are the predictable geometric ratios required to scale a rigid, 48-vertex Archimedean core (operating at a 3:4 Lissajous resonance) down to 4 dimensions without breaking its inherent symmetry. They represent the topological "friction" and scaling boundaries of the projection.

2. Curvature in M^4 as Topological Distortion

Standard Minkowski spacetime (M^4) is mathematically defined as flat. General Relativity introduces curvature to explain gravity, but does not explain the geometric *cause* of the bending. By conceptualizing the universe as a projection, curvature becomes the inevitable result of **projection stress**.

Just as projecting a 3D globe onto a 2D map inevitably causes distortion (stretching or compressing continents), projecting an enclosed 7-dimensional Torus into a 4D flat plane mandates localized topological deformations. The missing 3 dimensions must be accounted for geometrically; this compensation is what we observe as the bending of spacetime.